

# CF Tree: A Height-Balanced Tree Storing Clustering Features for Hierarchical Clustering

---

- ❑ Incremental insertion of new points (similar to B+-tree)
- ❑ For each point in the input
  - ❑ Find its closest leaf entry
  - ❑ Add point to leaf entry and update CF
  - ❑ If entry diameter  $>$  max\_diameter
    - ❑ Split leaf, and possibly parents
- ❑ A CF tree has two parameters
  - ❑ Branching factor: Maximum number of children
  - ❑ Maximum diameter of sub-clusters stored at the leaf nodes
- ❑ A CF tree: A height-balanced tree that stores the clustering features (CFs)
- ❑ The non-leaf nodes store sums of the CFs of their children

# CF Tree: A Height-Balanced Tree Storing Clustering Features for Hierarchical Clustering

